THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS UNIT TEST SET B

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. A rule that tells how to get each term of a sequence is called a: 1

(a) pattern rule **(b)** fraction **(c)** angle **(d)** perimeter

2. The sum of the first 15 odd numbers is: 1

(a) 240 **(b)** 210 **(c)** 30 **(d)** 225

3. The corners of a book form which type of angle? 1

(a) reflex angle **(b)** acute angle **(c)** right angle **(d)** obtuse angle

4. An angle measuring 120° is: 1

(a) reflex **(b)** obtuse **(c)** acute **(d)** right

5. Lines that meet at a right angle are: 1

(a) reflex lines **(b)** curved lines

(c) parallel lines **(d)** perpendicular lines

6. The number 1089 magic: 3-digit number minus its reverse, then add the reverse of the result. The final answer is: 1

(a) 1001 **(b)** 1089 **(c)** 999 **(d)** 1111

7. The largest 4-digit number from 3, 8, 1, 5 is: 1

(a) 8351 **(b)** 8513 **(c)** 5381 **(d)** 8531

8. ‘Mental math’ means: 1

(a) guessing

(b) calculating quickly in the mind using tricks

(c) only using a calculator

(d) using an abacus

9. A tally mark group of five is drawn as: 1

(a) five vertical lines

(b) a triangle

(c) four vertical lines crossed by a slanting line

(d) a circle

10. Data on favourite sports of 40 children could be best collected by: 1

(a) asking each child **(b)** measuring height

(c) guessing **(d)** counting classrooms

11. A number is divisible by 10 if its last digit is: 1

(a) 0 **(b)** 5 **(c)** 1 **(d)** any

12. The HCF of 12 and 18 is: 1

(a) 7 **(b)** 6 **(c)** 5 **(d)** 12

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. What is a sequence? Give one example. 2

14. Name the types of angles by their measures. 2

15. Round 7,468 to the nearest hundred and to the nearest thousand. 2

16. What is a pictograph? What is a key? 2

17. Write the first five multiples of 7. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 30–40 words each.

18. Explain why 1, 3, 6, 10, … are called triangular numbers. 3

19. How many degrees does the minute hand turn in 20 minutes? How about the hour hand in 20 minutes? 3

20. A pictograph uses one symbol = 10 books. Library issues 35, 50, 20 books on three days. How many symbols each day? 3

21. Find the prime factorisation of 84 and 126 and their HCF. 3

22. Explain the Sieve of Eratosthenes in steps. 3

OR

Find the LCM of 12 and 18 using prime factorisation. 3

23. Find the 20th term of the sequence 3, 6, 9, … and explain. 3

OR

Draw the next shape in a sequence of squares with 1, 4, 9 dots and say how many dots it has. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

24. (i) Explain what a shape sequence is. 5
(ii) Draw the first three shapes of a stacked-squares sequence.
(iii) Find the number of small squares in the 5th shape.

OR

Case study: A tile worker lays tiles in a triangle with 1 tile in the first row, 2 in the second, 3 in the third and so on. 5
(i) How many tiles in row 7?
(ii) How many tiles in the first 7 rows?
(iii) Which sequence does the total follow?
(iv) How many rows are needed for 36 tiles?

25. (i) What are palindromic numbers? 5
(ii) List all 3-digit palindromes with digit sum 8.
(iii) Find the sum of 247 and its reverse.

OR

Case study: A calendar shows dates in a 7-column grid. 5
(i) Pick a 2×2 box 8, 9, 15, 16 and find the sum of diagonals.
(ii) What do you notice?
(iii) Test with another box.
(iv) Explain why.

END OF QUESTION PAPER Total: 50 Marks